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Week 3

VERIFY INSERTED DOCUMENTS

ORGANIZATION: Manasvi's Org - 2026... PROJECT: Project 0

Compare tiers ? [Icons] M

Data Explorer

{ } My Queries

⚙ Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- BankDB
 - Customers
 - Transactions
- admin
- bookflow
 - book_metadata**
 - books
- local
- sample_mflix

book_metadata +

Cluster0 > bookflow > book_metadata

View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

25 1 - 5 of 5

	_id ObjectId	book_id Int32	title String	author S
1	ObjectId('6a7eb6a236c8a9b...	101	"The Martian"	"Andy We
2	ObjectId('6a7eb75b36c8a9b...	102	"Dune"	"Frank H
3	ObjectId('6a7eb76d36c8a9b...	103	"Clean Code"	"Robert
4	ObjectId('6a7eb77e36c8a9b...	104	"Artificial Intelligence ...	"John Sm
5	ObjectId('6a7eb79136c8a9b...	105	"Data Structures"	"Mark Al

COMPARISON OPERATOR \$gt

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Compare tiers ? [Icons] M

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Documents 5 Aggregations Schema Indexes 1 Validation

{ "reviews.rating": { "\$gt": 4 } } [Generate query](#) Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

	_id ObjectId	book_id Int32	title String	author String
1	ObjectId('6a7eb6a236c8a9b...	101	"The Martian"	"Andy Weir"
2	ObjectId('6a7eb75b36c8a9b...	102	"Dune"	"Frank Herbert"
3	ObjectId('6a7eb76d36c8a9b...	103	"Clean Code"	"Robert C. Martin"
4	ObjectId('6a7eb77e36c8a9b...	104	"Artificial Intelligence ...	"John Smith"
5	ObjectId('6a7eb79136c8a9b...	105	"Data Structures"	"Mark Allen"

USING \$IN

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Compare tiers ? [Menu] [User] [Bell] [Grid]

Data Explorer

My Queries Data Modeling

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book_metadata +

Cluster0 > bookflow > book_metadata

View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 1 Validation

```

{
  "reviews.rating": {
    "$in": [4, 5]
  }
}

```

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE 25 1-5 of 5

_id	ObjectId	book_id	Int32	title	String	author	String
1	ObjectId('6a7eb6a236c8a9b...')	101		"The Martian"		"Andy Weir"	
2	ObjectId('6a7eb75b36c8a9b...')	102		"Dune"		"Frank Herbert"	
3	ObjectId('6a7eb76d36c8a9b...')	103		"Clean Code"		"Robert C. Martin"	
4	ObjectId('6a7eb77e36c8a9b...')	104		"Artificial Intelligence ..."		"John Smith"	
5	ObjectId('6a7eb79136c8a9b...')	105		"Data Structures"		"Mark Allen"	

REGEX QUERY

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Data Explorer

My Queries Data Modeling

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book_metadata +

Cluster0 > bookflow > book_metadata

View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 1 Validation

```

{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}

```

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE 25 1-4 of 4

_id	ObjectId	book_id	Int32	title	String	author	String
1	ObjectId('6a7eb6a236c8a9b...')	101		"The Martian"		"Andy Weir"	
2	ObjectId('6a7eb75b36c8a9b...')	102		"Dune"		"Frank Herbert"	
3	ObjectId('6a7eb77e36c8a9b...')	104		"Artificial Intelligence ..."		"John Smith"	
4	ObjectId('6a7eb79136c8a9b...')	105		"Data Structures"		"Mark Allen"	

AGGREGATION PIPELINE

STEP 1: \$MATCH

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Compare tiers ? [Icons]

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Search clusters

book_metadata BankDB bookflow +

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate aggregation ? Explain Run Options

Untitled - modified SAVE + CREATE NEW EXPORT CODE PREVIEW STAGES TEXT WIZARD

5 Documents in the collection

Stage 1 \$match

```

1 - {
2
3   "published_year": {
4     "$gt": 2020
5   }
6 }
7 -

```

Output preview after \$match stage (Sample of 4 documents)

```

_id: ObjectId('6a7eb6a236c8a9b4f38f0908')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101

```

STEP 2: \$UNWIND

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Compare tiers ? [Icons]

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Search clusters

book_metadata BankDB bookflow +

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate aggregation ? Explain Run Options

Untitled - modified SAVE + CREATE NEW EXPORT CODE PREVIEW STAGES TEXT WIZARD

Stage 1 \$unwind

```

1 "$reviews"
2

```

Output preview after \$unwind stage (Sample of 10 documents)

```

_id: ObjectId('6a7eb6a236c8a9b4f38f0908')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101

```

STEP 3: \$GROUP

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Compare tiers

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book_metadata | BankDB | bookflow

Cluster0 > bookflow > book_metadata

Documents: 5 | Aggregations | Schema | Indexes: 1 | Validation

Sgroup

Generate aggregation Explain Run Options

Untitled - modified SAVE CREATE NEW EXPORT CODE PREVIEW STAGES TEXT WIZARD

> 5 Documents in the collection

Stage1: Sgroup

```
1 {
2   "_id": "title",
3   "average_rating": {
4     "$avg": "$reviews.rating"
5   },
6   "total_reviews": {
7     "$sum": 1
8   }
9 }
10
11
12
```

Output preview after Sgroup stage (Sample of 5 documents)

```
_id: "The Martian"
average_rating: null
total_reviews: 1
```

```
_id: "D
average_r
total_r
```

STEP 4: \$SORT

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Compare tiers

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book_metadata | BankDB | bookflow

Cluster0 > bookflow > book_metadata

Documents: 5 | Aggregations | Schema | Indexes: 1 | Validation

\$sort

Generate aggregation Explain Run Options

Untitled - modified SAVE CREATE NEW EXPORT CODE PREVIEW STAGES TEXT WIZARD

> 5 Documents in the collection

Stage1: \$sort

```
1 {
2   "average_rating": -1
3 }
4
5
6
```

Output preview after \$sort stage (Sample of 5 documents)

```
_id: ObjectId('6a7eb6a236c8a9b4f3870908')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
```

```
_id: 0b
book_id
title:
author:
publish
categor
format
file_si
member_
```

CREATE TEXT INDEX

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 - local
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book_metadata

BankDB

bookflow

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema **Indexes 2** Validation

Create Index Refresh

Looking for search indexes?
These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 kB	1 (since Mon Aug 24 2026)	UNIQUE	READY
> reviews.comments_text	TEXT	20.5 kB	0 (since Thu Aug 27 2026)		READY

SEARCH USING TEXT INDEX

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book_metadata

BankDB

bookflow

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

View monitoring Visualize Your Data

Generate query Explain Reset Find Options

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

ADD DATA BULK EXPORT CODE

25 1-5 of 5

book_metadata	_id ObjectId	book_id Int32	title String	author String
1	ObjectId('6a7eb5a236c8a9b...')	101	"The Martian"	"Andy Weir"
2	ObjectId('6a7eb76d36c8a9b...')	103	"Clean Code"	"Robert C. Martin"
3	ObjectId('6a7eb75b36c8a9b...')	102	"Dune"	"Frank Herbert"
4	ObjectId('6a7eb79136c8a9b...')	105	"Data Structures"	"Mark Allen"
5	ObjectId('6a7eb77e36c8a9b...')	104	"Artificial Intelligence ..."	"John Smith"

SEARCH FOR A SPECIFIC KEYWORD

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 - bookflow
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 - books

book_metadata

BankDB

bookflow

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

View monitoring Visualize Your Data

Generate query Explain Reset Find Options

```
{
  "$text": {
    "$search": "\"science fiction\""
  }
}
```

ADD DATA BULK EXPORT CODE

25 1-2 of 2

book_metadata	_id ObjectId	book_id Int32	title String	author String
1	ObjectId('6a7eb75b36c8a9b...')	102	"Dune"	"Frank Herbert"
2	ObjectId('6a7eb6a236c8a9b...')	101	"The Martian"	"Andy Weir"

USE explain()

The screenshot shows the MongoDB Atlas Data Explorer interface. The left sidebar contains the 'Data Explorer' section with a tree view of clusters and databases. The main panel displays the 'book_metadata' collection in the 'bookflow' database. A JSON document is shown in the top pane, and a table view of the collection is shown in the bottom pane. The table has columns: _id, ObjectId, book_id, Int32, title, String, and author, String. The table contains 5 rows of data.

	_id	ObjectId	book_id	Int32	title	String	author	String
1	ObjectId('6a7eb6a236c8a9b...	101			"The Martian"		"Andy Weir"	
2	ObjectId('6a7eb76d36c8a9b...	103			"Clean Code"		"Robert C. Martin"	
3	ObjectId('6a7eb75b36c8a9b...	102			"Dune"		"Frank Herbert"	
4	ObjectId('6a7eb79136c8a9b...	105			"Data Structures"		"Mark Allen"	
5	ObjectId('6a7eb77e36c8a9b...	104			"Artificial Intelligence ..."		"John Smith"	